THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** cell	Cell.
2	**(a)** similar cells	Same function.
3	**(b)** variety of life	Diversity.
4	**(a)** making new individuals	Continuity.
5	**(d)** energy	Main source.
6	**(b)** water	Through a membrane.
7	**(b)** epithelial tissue	Covering.
8	**(d)** Porifera	Pores.
9	**(b)** ovary	After fertilisation.
10	**(d)** much sunlight	High albedo.
11	**(b)** two identical cells	Growth.
12	**(c)** hinge joint	One plane.
13	**(a)** Reptilia	Scales.
14	**(c)** testes	Male.
15	**(b)** carbon dioxide	CO_2.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Basic structural and functional unit of life (2).	2
17	Group of similar cells performing a function (2).	2
18	Grouping of organisms based on similarities (2).	2
19	Production of new individuals of the same kind (2).	2
20	Geosphere and hydrosphere (1); Atmosphere and biosphere (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Cell wall (1); Plastids (1); Vacuole size (1).	1+1+1
22	Apical (1); Lateral (1); Intercalary (1).	1+1+1
23	Cell type (1); Body organisation (1); Nutrition (1).	1+1+1
24	Fission: Amoeba (1); Budding: Hydra (1); Spores: bread mould (1).	1+1+1
	OR: Fossil fuels (1); Deforestation (1); Agriculture and waste (1).	1+1+1
25	Protects (1); Controls entry (1); Controls exit (1).	1+1+1
	OR: Transport (1); Defence (1); Clotting (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Membrane (1); Nucleus (1); Cytoplasm (1); Organelles (1); Functions (1).	1+1+1+1+1
	OR: Definition (1); Apical (1); Lateral (1); Intercalary (1); Features (1).	1+1+1+1+1
27	Monera (1); Protista (1); Fungi (1); Plantae (1); Animalia (1).	1+1+1+1+1
	OR: Definition (1); Fission (1); Budding (1); Spores (1); Vegetative (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** Robert Hooke	1665.
2	**(c)** meristematic	Cell division.
3	**(c)** similarities and differences	Characters.
4	**(c)** binary fission	Two parts.
5	**(a)** atmosphere	Gases.
6	**(a)** high to low concentration	Gradient.
7	**(a)** nervous	Neurons.
8	**(a)** Cnidaria	Stinging cells.
9	**(b)** pollination	Pollination.
10	**(a)** absorb sunlight	Low albedo.
11	**(b)** gametes	Half chromosomes.
12	**(c)** ball and socket joint	All directions.
13	**(d)** Aves	Feathers.
14	**(d)** ovaries	Female.
15	**(b)** carbon dioxide	CO_2.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Cells without a nucleus (1); Bacteria (1).	1+1
17	Conducting tissue carrying food (2).	2
18	Group of organisms that can interbreed (2).	2
19	Transfer of pollen from anther to stigma (2).	2
20	Change of water into vapour (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	No vascular tissue (1); Need water (1); Small (1).	1+1+1
22	Testes (1); Vas deferens (1); Penis (1).	1+1+1
23	Save water (1); Save energy (1); Avoid plastics (1).	1+1+1
24	Nucleus (1); Mitochondria (1); Chloroplasts (1).	1+1+1
	OR: Receive messages (1); Carry messages (1); Coordinate (1).	1+1+1
25	Gills (1); Fins (1); Scales (1).	1+1+1
	OR: Small (1); Light pollen (1); Feathery stigma (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Geosphere (1); Hydrosphere (1); Atmosphere (1); Biosphere (1); Interactions (1).	1+1+1+1+1
	OR: Process (1); Two cells (1); Growth (1); Repair (1); Same chromosomes (1).	1+1+1+1+1
27	Neuron (1); Dendrites (1); Axon (1); Impulse (1); Function (1).	1+1+1+1+1
	OR: Cotyledons (1); Roots (1); Leaves (1); Flowers (1); Examples (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** nucleus	Contains DNA.
2	**(b)** xylem	Upward.
3	**(c)** binomial nomenclature	Naming.
4	**(c)** budding	Buds.
5	**(b)** geosphere	Solid.
6	**(b)** swells	Water enters.
7	**(c)** muscular	Contraction.
8	**(c)** flatworm	Flat.
9	**(c)** fertilisation	Zygote.
10	**(d)** hotter than the poles	Direct rays.
11	**(d)** 46	23 pairs.
12	**(a)** fixed joints	No movement.
13	**(c)** Mammalia	Hair and milk.
14	**(b)** oviduct	Fallopian tube.
15	**(c)** more CO_2 in air	Fewer trees.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Photosynthesis (2).	2
17	Tissue connecting bone to bone (2).	2
18	Porifera (1); Arthropoda (1).	1+1
19	Testes (1); Ovaries (1).	1+1
20	Fraction of sunlight reflected by a surface (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Absorb CO_2 (1); Give oxygen (1); Rain (1).	1+1+1
22	Nucleus (1); Mitochondria (1); Ribosomes (1).	1+1+1
23	Parenchyma (1); Collenchyma (1); Sclerenchyma (1).	1+1+1
24	Kingdom, phylum, class (1); Order, family (1); Genus, species (1).	1+1+1
	OR: Sepals (1); Petals (1); Stamens and carpels (1).	1+1+1
25	Melting ice (1); Rising sea level (1); Extreme weather (1).	1+1+1
	OR: Control centre (1); Contains DNA (1); Heredity (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Barrier (1); Chemical (1); IUD (1); Surgical (1); Uses (1).	1+1+1+1+1
	OR: Source (1); Speed (1); Distribution (1); Reflection (1); Use (1).	1+1+1+1+1
27	Wall (1); Membrane (1); Plastids (1); Vacuole (1); Nucleus (1).	1+1+1+1+1
	OR: Parenchyma (1); Collenchyma (1); Sclerenchyma (1); Features (1); Functions (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.